

Ans. Shed 1 ML

Q1 Find gmin

$$1- f(x) = 3x^2 + 82x - 9.91$$

$$\frac{df(x)}{dx} = 3 \times 2x + 82$$

$$\frac{df(x)}{dx} = 0$$

$$6x = -82$$

$$x = -\frac{82}{6}$$

$$y = 3\left(\frac{-82}{6}\right)^2 + 82 \times \frac{-82}{6} - 9.91$$

$$= -10.24$$

$$\left(\frac{-82}{6}, -10.24\right)$$

$$f(x) = \frac{1}{3}x^3 + 2x^2 + 2x + 3$$

$$f'(x) = x^2 + 4x + 2 = 0$$

$$b^2 - 4ac < 0$$

$\therefore f(x)$ doesn't have global minimum

المسح الضوئي

Q2) $f(x) = 3x^2 + 3x - 1$

$$f'(x) = 6x + 3$$

Iteration (1) $x_0 = -1$

$$f(x_0) = -1$$

$$f'(x_0) = 3$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_1 = -1 + \frac{1}{3} = -\frac{2}{3}$$

$$f\left(-\frac{2}{3}\right) = -\frac{1}{3}$$

$$f'\left(-\frac{2}{3}\right) = 5$$

$$x_2 = -\frac{2}{3} - \frac{-\frac{1}{3}}{5} = -\frac{1}{5} = -0.2$$

$$f(-0.2) = 0.0134$$

$$f'(-0.2) = 4.6002$$

$$x_3 = -0.2 - \frac{0.0134}{4.6002} = -0.2291$$

$$x_0 = -1$$

$$f(-1) = -1$$

$$f'(-1) = -3$$

$$x_1 = -\frac{4}{3}$$

$$f\left(-\frac{4}{3}\right) = \frac{1}{3}$$

$$f'\left(-\frac{4}{3}\right) = -5.2667$$

$$x_2 = -1.2667$$

$$f(-1.2667) = 0.0134$$

$$f'(-1.2667) = -4.6002$$

$$x_3 = -1.2638$$

The second root

$$f(-1.2638) = 0.0001 \approx 0$$

The first root

$$P(x) = \frac{1}{3}x^3 + 2x^2 + 2x + 3$$

$$P'(x) = x^2 + 4x + 2$$

$$P(-5) = 1.33 \quad P(-6) = -9$$

$$P(x) = 0$$

$$-5 < x < -6$$

$$x_0 = -5$$

$$P(-5) = 1.333$$

$$P'(-5) = 7$$

$$x_1 = -5.1905$$

$$P(-5.1905) = -0.1141$$

$$P'(-5.1905) = 8.1793$$

$$x_2 = -5.1905 - \frac{-0.1141}{8.1793}$$

$$x_3 = -5.175 \rightarrow \text{النتيجة هي القيمة}$$

~~x = -6~~
The second, third root
in complex number

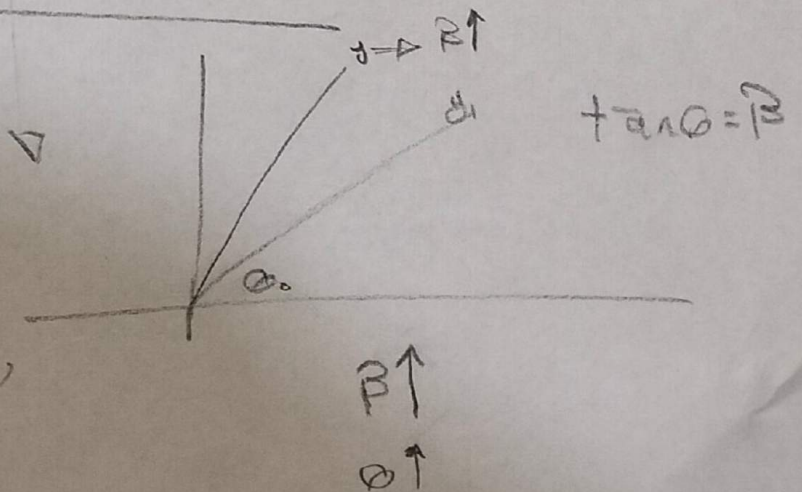
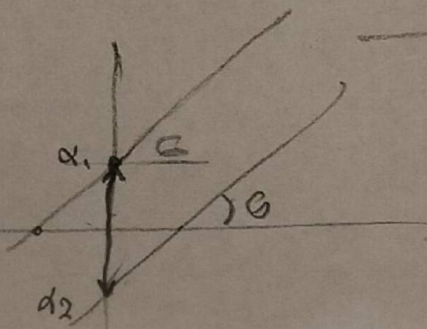
$$[4] \quad f(x) = \alpha + \beta x$$

$$\alpha = 0.5$$

$$\alpha_1 = *$$

$$\alpha_2 = -1$$

$\alpha \downarrow$
shift in α



Q5

X Matrix

$$\begin{bmatrix} 4 & 220 & 1 \\ 3 & 190 & 1 \\ 3 & 150 & 1 \\ 2 & 140 & 1 \\ 3 & 110 & 1 \end{bmatrix}$$

Y Matrix

$$\begin{bmatrix} 300 \\ 250 \\ 220 \\ 180 \\ 200 \end{bmatrix}$$

Correlation

$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$\bar{x} = \frac{15}{5} = 3$$

$$\bar{y} = \frac{1150}{5} = 230$$

	x_1 N. of room	x_2 area	y Price
x_1 n. of room	1	,6541	,9045
x_2 area	,6541	1	,9121
Price	,9045	,9121	1

$$C_{11}(x_1, x_1) = ,6541$$

$$C_{12}(x_1, y) = ,9045$$

$$C_{22}(x_2, y) = ,9121$$

~~MAE~~ Mean Absolute Error MAE

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i| = 230$$

Learning rate

$$w_{new} = w_0 + \alpha \left(\frac{1}{n} x^T (\bar{y} - y) \right)$$

Learning rate

mean

$$\frac{\partial w}{\partial y} =$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} -230 \\ -714 \\ -38740 \end{bmatrix} = \checkmark$$

$$w_3 = \frac{1}{5} \sum (-y_i \alpha) = -38740$$

$$w_1 * w_2 x_1 + w_3 x_2 = y$$

$$w_1 = \frac{1}{n} \sum (-y) = -230$$

$$w_2 = \frac{1}{5} \sum (-y \cdot \text{room}) = -714$$

$$\omega_{\text{new}} = \omega_{\text{old}} - \alpha \left[\frac{\partial \omega}{\partial \theta} \right] \rightarrow \frac{\partial}{\partial \theta}$$

$$y = \frac{1}{2M} \sum h(x(i)) - y(i)$$

$$\frac{\partial \omega}{\partial \theta} = \frac{1}{2M} \sum_{i=1}^M \frac{h - y}{n - y} x_i$$

$$\omega_{\text{new}} = \omega_{\text{old}} - \alpha \frac{h - y}{2M |n - y|} x_i$$

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